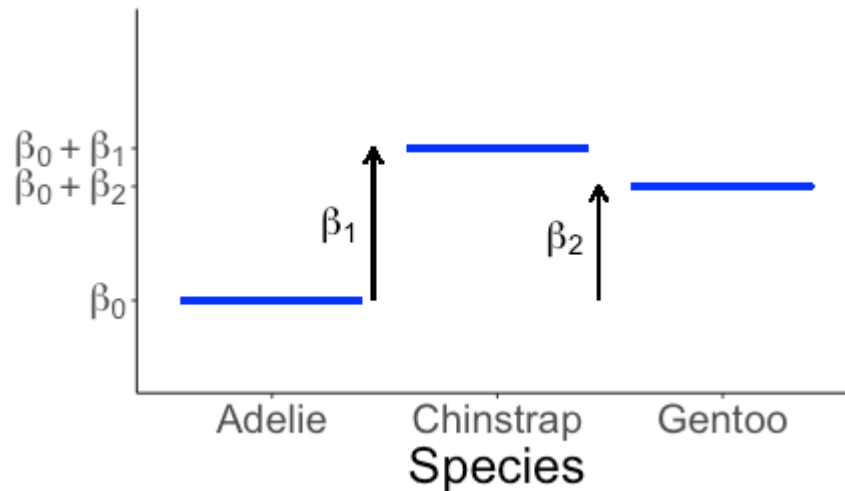


Regression with a categorical predictor

Reminders and agenda

- + Project part 2 due tonight
- + HW 4 due Monday
- + Today:
 - + Continue thinking about categorical explanatory variables
 - + How do we summarize strength of a relationship when there are more terms in the model?

Previously: regression with a categorical predictor

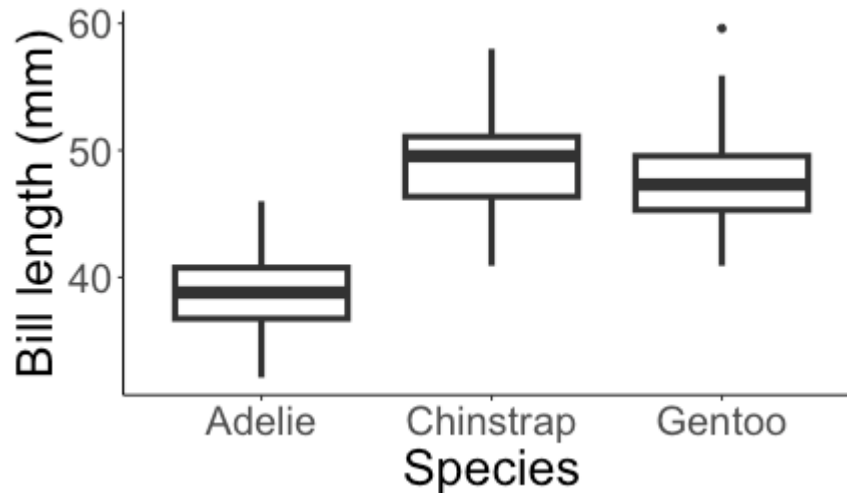


$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Or, more concisely:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

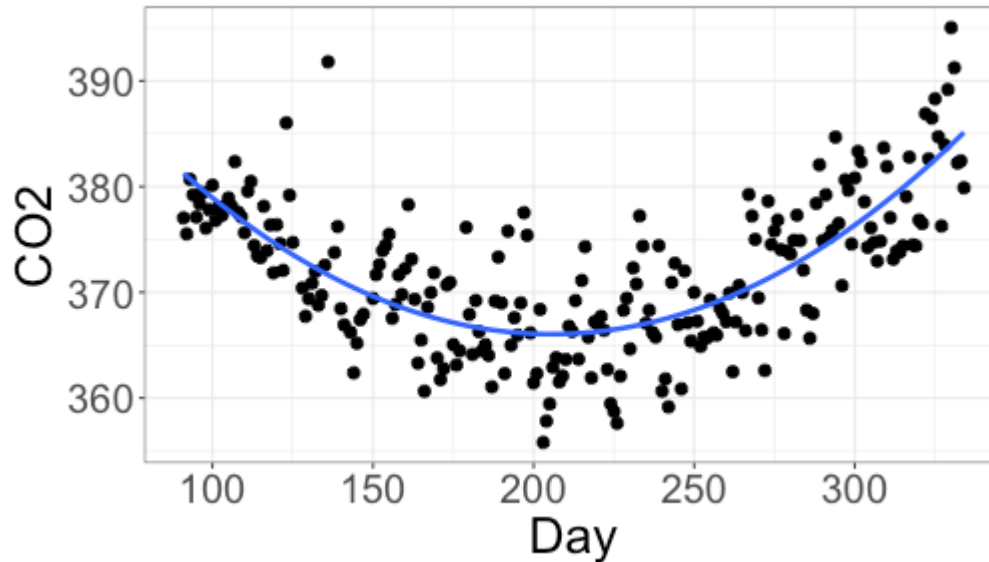
Previously: regression with a categorical predictor



Can I use the correlation to summarize the relationship between species and bill length?

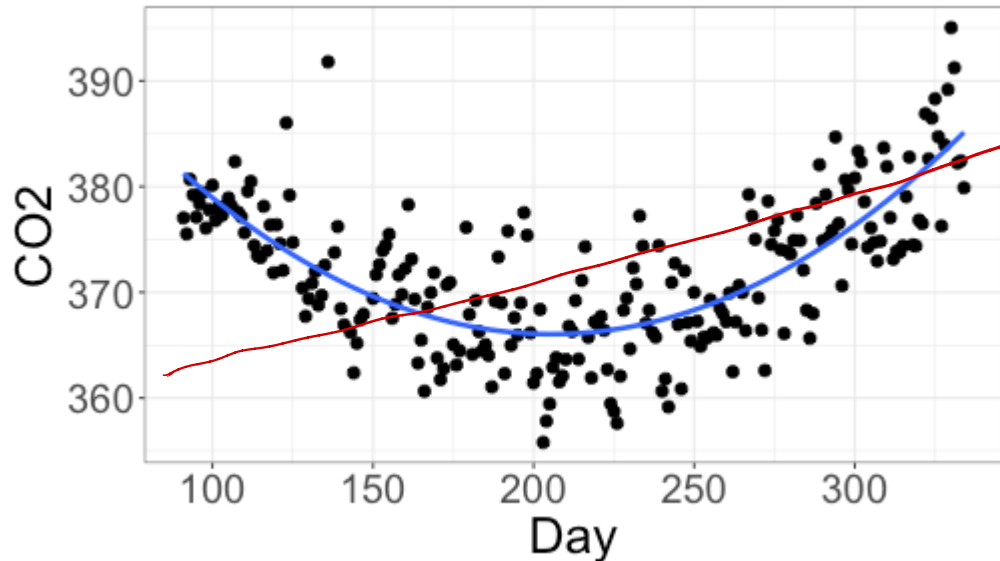
No! Correlation is for 2 quantitative variables

Previously: polynomial transformations



$$\widehat{\text{CO2}} = 414.975 - 0.476 \text{ Day} + 0.0012 \text{ Day}^2$$

Previously: polynomial transformations



$$\widehat{\text{CO2}} = 414.975 - 0.476 \text{ Day} + 0.0012 \text{ Day}^2$$

Can I use the correlation to calculate the relationship between Day and CO2?

• can calculate correlation
• correlation is not a good description with nonlinear relationships 6/22

Measuring variability in the response

- When we fit a regression model with least squares, we choose parameters to minimize the sum of squared residuals:

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

observed response prediction from fitted model

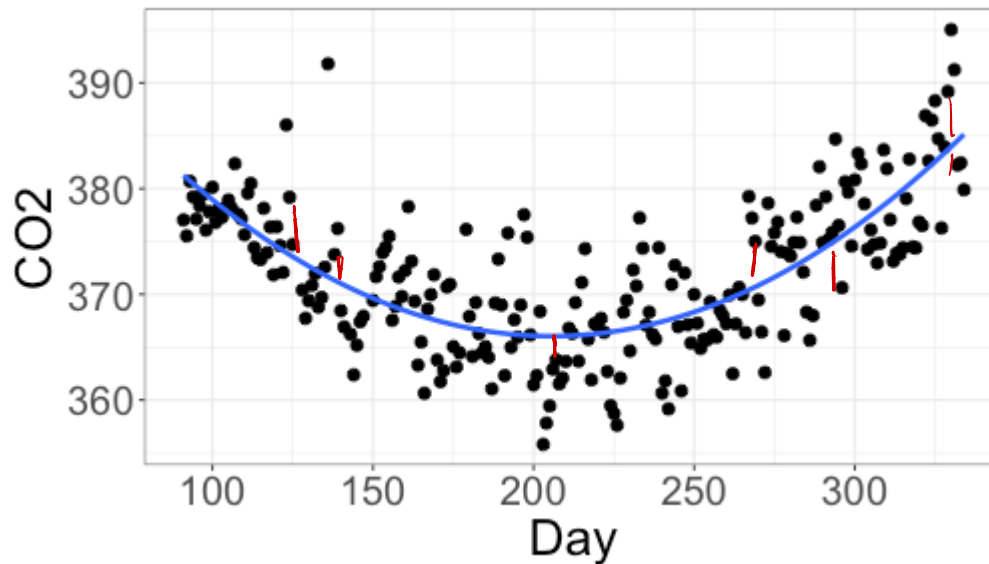
- We can think of SSE as the "amount of variability" in Y after predicting with our regression model
- The amount of variability in Y *before* predicting with our regression model is

$$SSTotal = \sum_i (y_i - \bar{y})^2$$

(total sum of squares) overall mean of the observed y values (no model fit)

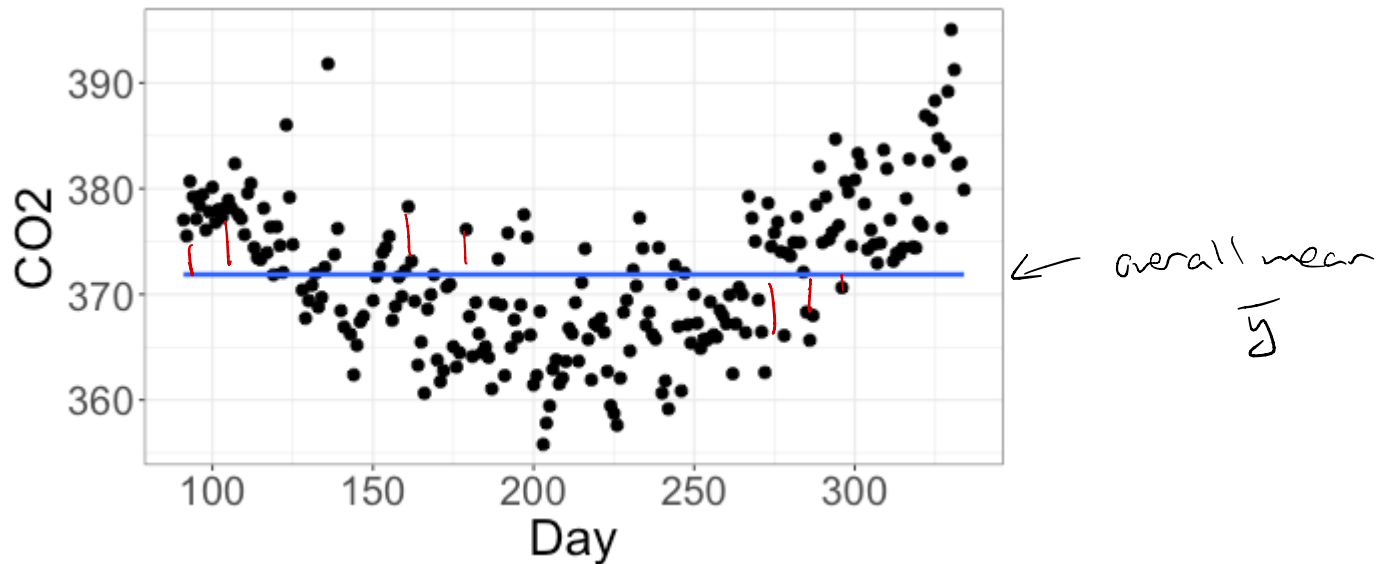
Thinking about SSE

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

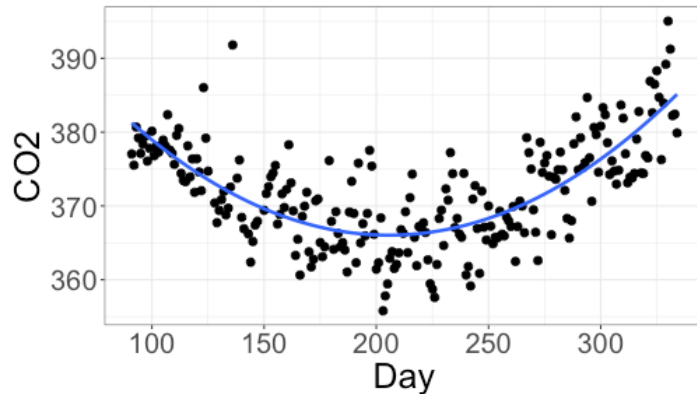


Thinking about SS_{Total}

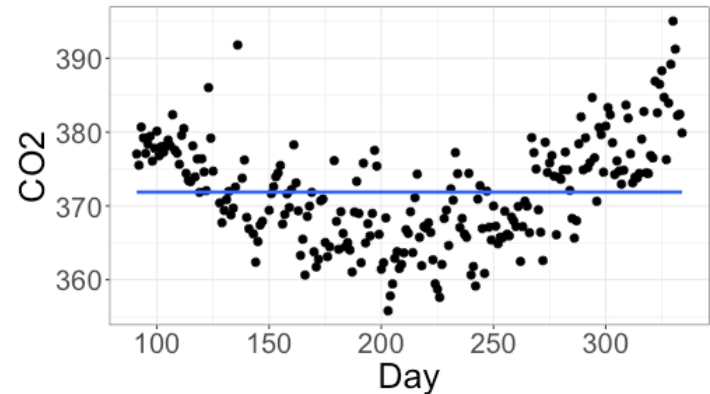
$$SS_{Total} = \sum_i (y_i - \bar{y})^2$$



SSE vs. SSTotal



SSE = 4992.71



SSTotal: 11703.69

Do I want SSE to be large relative to SSTotal, or small relative to SSTotal?

want SSE to be smaller (much smaller)
than SSTotal

R^2 : the coefficient of determination

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

$$SSE \leq SSTotal$$

$$SSTotal = \sum_i (y_i - \bar{y})^2 \Rightarrow \frac{SSE}{SSTotal} \leq 1$$

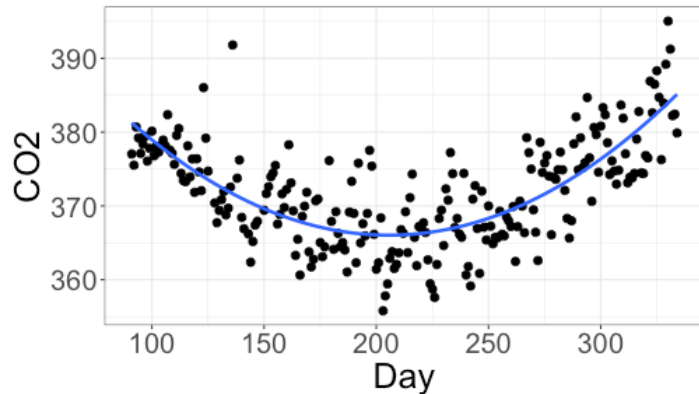
$$1 \geq 1 - SSE/SSTotal \geq 0$$

The proportion of variability in Y "explained" by our regression model is

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

- + R^2 is between 0 and 1
 - + A higher R^2 means our model is a "better fit" to the data
 - + For regression with one variable, $R^2 = r^2$ (the squared correlation)
(quantitative)
- bigger change and between SSTotal*

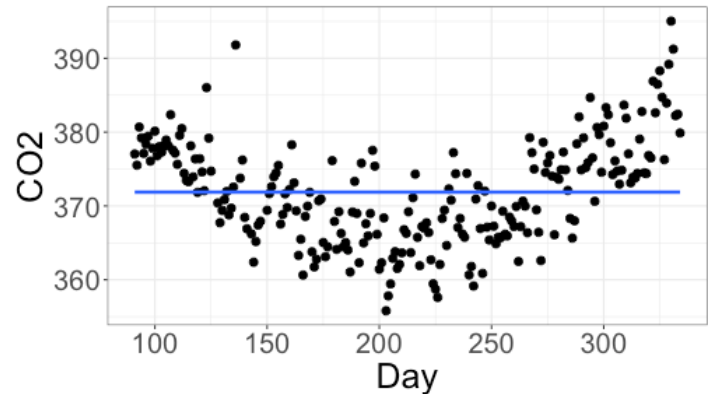
SSE vs. SSTotal



SSE = 4992.71

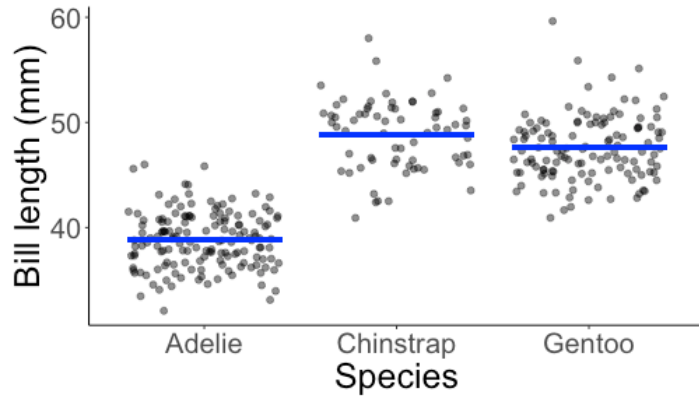
$$R^2 = 1 - \frac{4992.71}{11703.69} = 0.573$$

About 57.3% of the variability in CO2 levels is explained by our model with Day



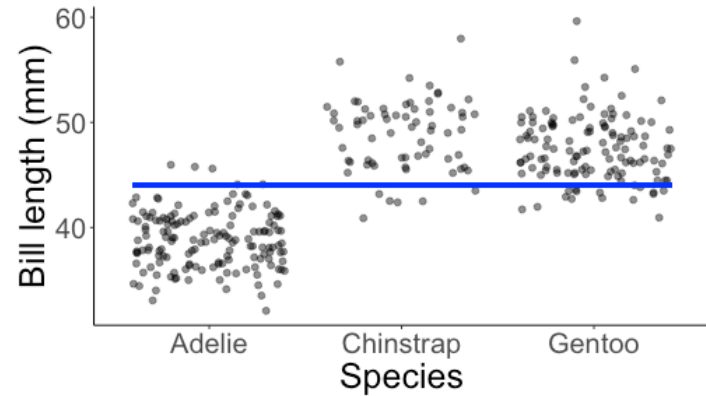
SSTotal: 11703.69

SSE vs. SSTotal



SSE = 2969.9

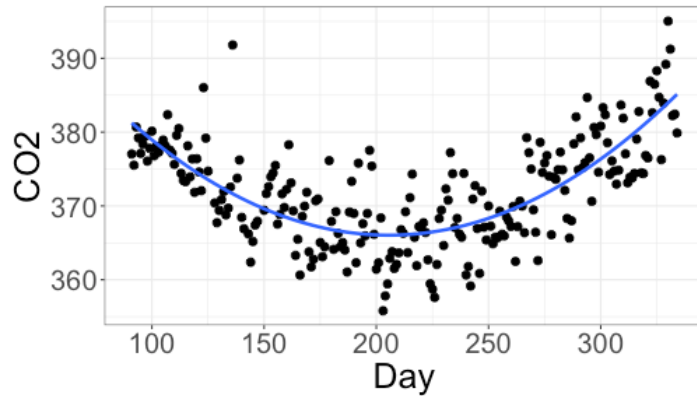
$$R^2 = 1 - \frac{2969.9}{10164.21} = 0.708$$



SSTotal = 10164.21

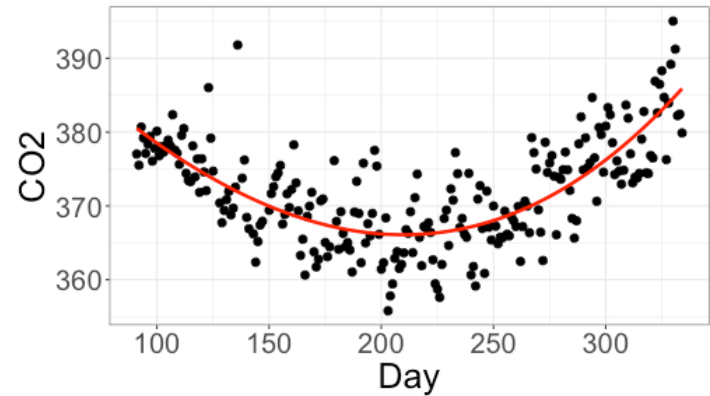
Comparing R^2 values

Polynomial of order 2:



$$R^2 = 0.573$$

Polynomial of order 3:



$$R^2 = 0.575$$

R^2 has increased very slightly -- does that mean that a polynomial of order 3 is a better fit?

No, the values are very close, and the polynomial of order 3 is more complex

R^2

R^2 will always increase when you add more parameters to the model.

Why? Because

$$R^2 = 1 - \frac{SSE}{SSTotal}$$

and adding more parameters gives us more flexibility to decrease SSE.

Adjusted R^2

We can account for the number of parameters in the model with *adjusted R^2* :

$$R_{adj}^2 = 1 - \frac{SSE/(n-p)}{SSTotal/(n-1)}$$

← penalize the number of parameters in the model

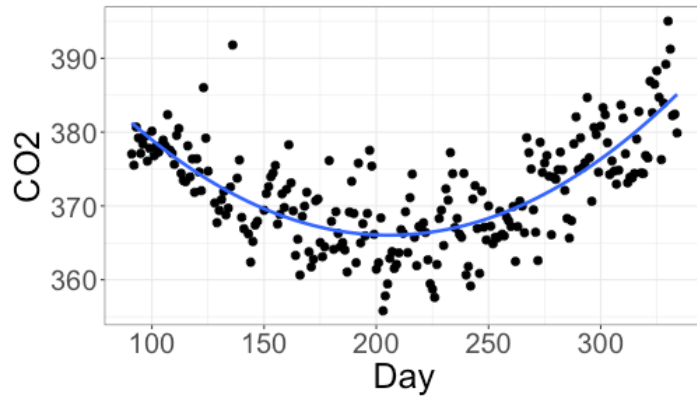
- + n = number of observations in data (sample size)
- + p = number of parameters (β terms) in model

$n-p$ = df for fitted model (lost a df for each β estimated)

$n-1$ = df for model with no explanatory variables (lost a df by estimating mean of y with $\bar{y}$)

Adjusted R^2

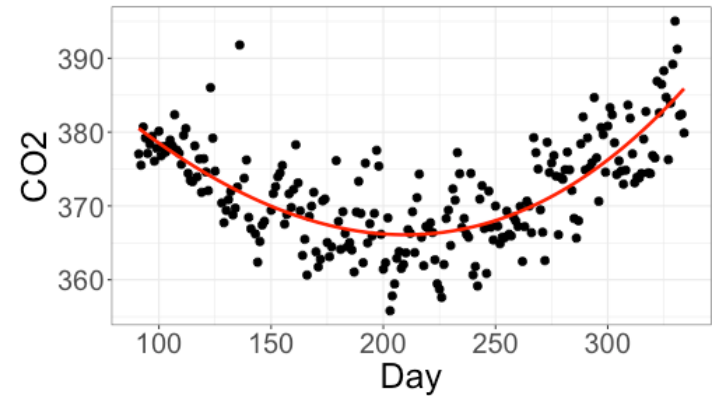
Polynomial of order 2:



SSE = 4992.7

SSTotal = 11703.69

Polynomial of order 3:



SSE = 4971.9

SSTotal = 11703.69

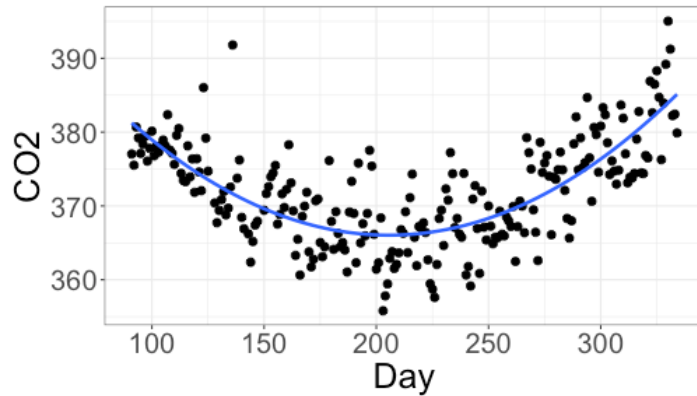
Why is SSTotal the same for the two models?

$$SSTotal = \sum_i (y_i - \bar{y})^2$$

Same $\bar{y}$ (Same response variable, same data) 17/22

Adjusted R^2

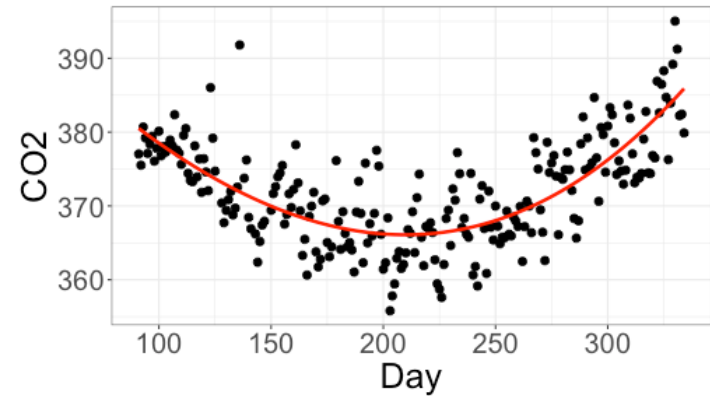
Polynomial of order 2:



SSE = 4992.7

SSTotal = 11703.69

Polynomial of order 3:



SSE = 4971.9

SSTotal = 11703.69

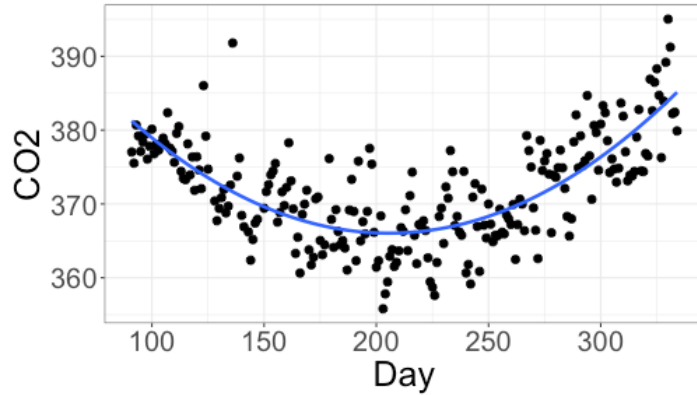
$$R_{adj}^2 = 1 - \frac{SSE/(n-p)}{SSTotal/(n-1)}$$

$$R_{adj}^2 = 1 - \frac{4992.7/(n-3)}{11703.69/(n-1)}$$

$$R_{adj}^2 = 1 - \frac{4971.9/(n-4)}{11703.69/(n-1)}$$

Adjusted R^2

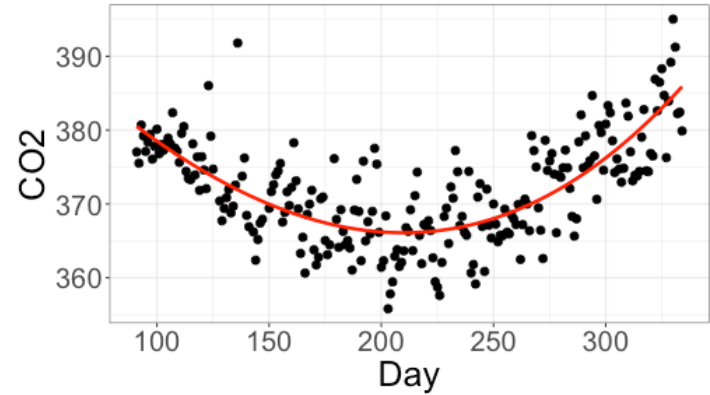
Polynomial of order 2:



$$R_{adj}^2 = 0.56976$$

$$R^2 = 0.573$$

Polynomial of order 3:



$$R_{adj}^2 = 0.56972$$

$$R^2 = 0.575$$

$$R_{adj}^2 < R^2$$

Properties of R^2_{adj}

- + $R^2_{adj} < R^2$ (always)
- + R^2_{adj} accounts for the number of parameters in the model
- + R^2 can *never* decrease when predictors are added
- + R^2_{adj} *can* decrease if unhelpful predictors are added

R^2 : summarizes how much variability in response is "explained" by the model)

R^2_{adj} : accounts for # parameters, which is useful for comparing different models

Values in R

```
penguin_mod <- lm(bill_length_mm ~ species,  
                  data = penguins)  
summary(penguin_mod)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	38.7914	0.2409	161.05	<2e-16	***
speciesChinstrap	10.0424	0.4323	23.23	<2e-16	***
speciesGentoo	8.7135	0.3595	24.24	<2e-16	***
...					
Multiple R-squared:	0.7078				
Adjusted R-squared:	0.7061				

R^2 R^2_{adj}

```
co2_mod <- lm(CO2 ~ poly(Day, 2, raw=TRUE),  
              data = CO2Germany)  
summary(co2_mod)
```

...

Multiple R-squared: 0.5734, Adjusted R-squared: 0.5698

Class activity

- + Practice regression with a single categorical predictor, with diamonds data
- + Calculate R^2 and R^2_{adj}
- + No need to submit this one on Canvas

https://sta112-f25.github.io/class_activities/ca_16.html